

```

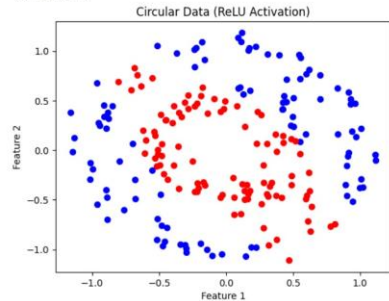
from sklearn.datasets import make_circles; from sklearn.neural_network import MLPClassifier
import matplotlib.pyplot as plt
X, y = make_circles(n_samples=200, noise=0.1, factor=0.5, random_state=42)
model = MLPClassifier(hidden_layer_sizes=(10,), activation='relu', max_iter=1000).fit(X, y)
plt.scatter(X[:,0], X[:,1], c=model.predict(X), cmap='bwr')
plt.title("Circular Data (ReLU Activation)")
plt.xlabel("Feature 1"); plt.ylabel("Feature 2")
plt.show()

```

```

... /usr/local/lib/python3.12/dist-packages/sklearn/neural_network/_multilayer_perceptron.py:
warnings.warn(

```



```

from sklearn.datasets import make_classification; from sklearn.neural_network import MLPClassifier
import matplotlib.pyplot as plt
X, y = make_classification(n_samples=200, n_features=2, n_redundant=0, n_clusters_per_class=1, random_state=42)
model = MLPClassifier(hidden_layer_sizes=(10,), activation='relu', max_iter=1000, random_state=42).fit(X, y)
plt.scatter(X[:,0], X[:,1], c=model.predict(X), cmap='bwr')
plt.title("Two-Class Data (ReLU Activation)")
plt.xlabel("Feature 1"); plt.ylabel("Feature 2")
plt.show()

```



```

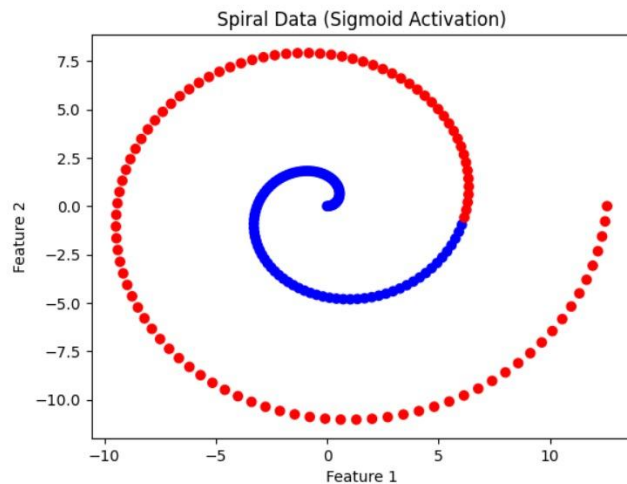
import numpy as np, matplotlib.pyplot as plt
from sklearn.neural_network import MLPClassifier
t = np.linspace(0,4*np.pi,200); X = np.c_[t*np.cos(t), t*np.sin(t)]; y = (t>2*np.pi).astype(int)
model = MLPClassifier(hidden_layer_sizes=(10,), activation='logistic', max_iter=1000, random_state=42).fit(X, y)
plt.scatter(X[:,0], X[:,1], c=model.predict(X), cmap='bwr')
plt.title("Spiral Data (Sigmoid Activation)")
plt.xlabel("Feature 1"); plt.ylabel("Feature 2"); plt.show()

```

```

... /usr/local/lib/python3.12/dist-packages/sklearn/neural_network/_multilayer_perceptron.py:
warnings.warn(

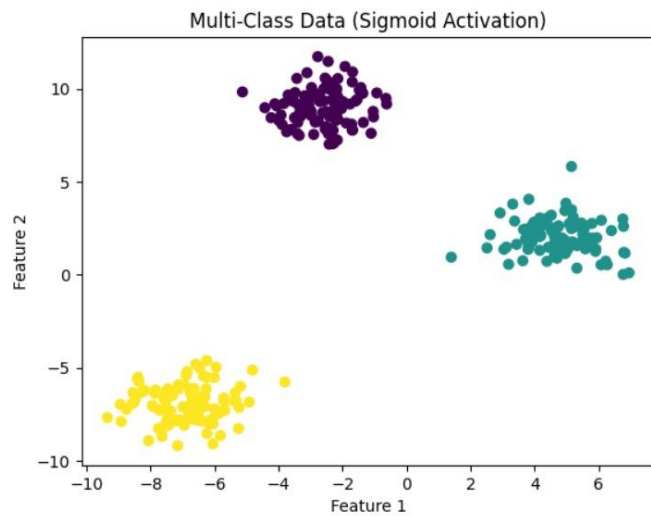
```



```

from sklearn.datasets import make_blobs; from sklearn.neural_network import MLPClassifier
import matplotlib.pyplot as plt
X, y = make_blobs(n_samples=300, centers=3, n_features=2, random_state=42)
model = MLPClassifier(hidden_layer_sizes=(10,), activation='logistic', max_iter=1000, random_state=42).fit(X, y)
plt.scatter(X[:,0], X[:,1], c=model.predict(X), cmap='viridis')
plt.title("Multi-Class Data (Sigmoid Activation)")
plt.xlabel("Feature 1"); plt.ylabel("Feature 2")
plt.show()

```



```

from sklearn.datasets import make_classification; from sklearn.neural_network import MLPClassifier
import matplotlib.pyplot as plt
X, y = make_classification(n_samples=200, n_features=2, n_redundant=0, n_clusters_per_class=1, random_state=42)
model = MLPClassifier(hidden_layer_sizes=(10,), activation='relu', learning_rate_init=0.001, max_iter=1000, random_state=42).fit(X, y)
plt.scatter(X[:,0], X[:,1], c=model.predict(X), cmap='bwr')
plt.title("Two-Class Data (ReLU, Learning Rate = 0.001)")
plt.xlabel("Feature 1"); plt.ylabel("Feature 2")
plt.show()

```

